

ACTIVATION-LEVEL UNCERTAINTY IN DEEP NEURAL NETWORKS

Pablo Morales-Álvarez*

Department of Computer Science and AI
University of Granada, Spain
pabloramoraless@decsai.ugr.es

Daniel Hernández-Lobato

Department of Computer Science
Universidad Autónoma de Madrid, Spain

Rafael Molina

Department of Computer Science and AI
University of Granada, Spain

José Miguel Hernández-Lobato

Department of Engineering
University of Cambridge, UK
Alan Turing Institute, London, UK

ABSTRACT

Current approaches for uncertainty estimation in deep learning often produce too confident results. Bayesian Neural Networks (BNNs) model uncertainty in the space of weights, which is usually high-dimensional and limits the quality of variational approximations. The more recent functional BNNs (fBNNs) address this only partially because, although the prior is specified in the space of functions, the posterior approximation is still defined in terms of stochastic weights. In this work we propose to move uncertainty from the weights (which are deterministic) to the activation function. Specifically, the activations are modelled with simple 1D Gaussian Processes (GP), for which a triangular kernel inspired by the ReLU non-linearity is explored. Our experiments show that activation-level stochasticity provides more reliable uncertainty estimates than BNN and fBNN, whereas it performs competitively in standard prediction tasks. We also study the connection with deep GPs, both theoretically and empirically. More precisely, we show that activation-level uncertainty requires fewer inducing points and is better suited for deep architectures.

1 INTRODUCTION

Deep Neural Networks (DNNs) have achieved state-of-the-art performance in many different tasks, such as speech recognition (Hinton et al., 2012), natural language processing (Mikolov et al., 2013) or computer vision (Krizhevsky et al., 2012). In spite of their predictive power, DNNs are limited in terms of uncertainty estimation. This has been a classical concern in the field (MacKay, 1992; Hinton & Van Camp, 1993; Barber & Bishop, 1998), which has attracted a lot of attention in the last years (Lakshminarayanan et al., 2017; Guo et al., 2017; Sun et al., 2019; Wenzel et al., 2020). Indeed, this ability to “know what is not known” is essential for critical applications such as medical diagnosis (Esteva et al., 2017; Mobiny et al., 2019) or autonomous driving (Kendall & Gal, 2017; Gal, 2016).

Bayesian Neural Networks (BNNs) address this problem through a Bayesian treatment of the network weights¹ (MacKay, 1992; Neal, 1995). This will be referred to as *weight-space* stochasticity. However, dealing with uncertainty in weight space is challenging, since it contains many symmetries and is highly dimensional (Wenzel et al., 2020; Sun et al., 2019; Snoek et al., 2019; Fort et al., 2019). Here we focus on two specific limitations. First, it has been recently shown that BNNs with well-established inference methods such as Bayes by Backprop (BBP) (Blundell et al., 2015) and MC-Dropout (Gal & Ghahramani, 2016) underestimate the predictive uncertainty for instances located in-between two clusters of training points (Foong et al., 2020; 2019; Yao et al., 2019). Second, the weight-space prior does not allow BNNs to guide extrapolation to out-of-distribution (OOD) data (Sun et al., 2019; Nguyen et al., 2015; Ren et al., 2019). Both aspects are illustrated graphically in Figure 3, more details in Section 3.1.

*Work developed mostly while visiting Cambridge University, UK.

¹The bias term will be absorbed within the weights throughout the work.

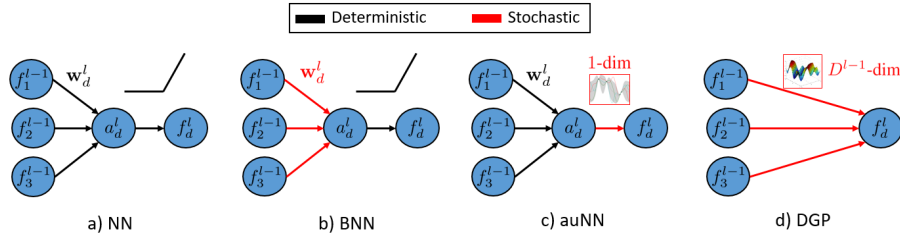


Figure 1: Graphical representation of the artificial neurons for closely related methods. The subscript d and the superscript l refer to the d -th unit in the l -th layer, respectively. (a) In standard Neural Networks (NN), both the weights and the activation function are deterministic. (b) In Bayesian NNs, weights are stochastic and the activation is deterministic. (c) In auNN (this work), weights are deterministic and the activation is stochastic. (d) Deep GPs do not have a linear projection through weights, and the output is modelled directly with a GP defined on the D^{l-1} -dimensional input space.

As an alternative to standard BNNs, Functional Bayesian Neural Nets (fBNN) specify the prior and perform inference directly in function space (Sun et al., 2019). This provides a mechanism to guide the extrapolation in OOD data, e.g. predictions can be encouraged to revert to the prior in regions of no observed data. However, the posterior stochastic process is still defined by a factorized Gaussian on the network weights (i.e. as in BBP), see (Sun et al., 2019, Sect. 3.1). We will show that this makes fBNN inherit the problem of underestimating the predictive uncertainty for in-between data.

In this work, we adopt a different approach by moving stochasticity from the weights to the activation function, see Figure 1. This will be referred to as auNN (activation-level uncertainty for Neural Networks). The activation functions are modelled with (one-dimensional) GP priors, for which a triangular kernel inspired by the ReLU non-linearity (Nair & Hinton, 2010; Glorot et al., 2011) is used. Since non-linearities are typically simple functions (e.g. ReLU, sigmoid, tanh), our GPs are sparsified with few inducing points. The network weights are deterministic parameters which are estimated to maximize the marginal likelihood of the model. The motivation behind auNN is to avoid inference in the complex space of weights. We hypothesise that it could be enough to introduce stochasticity in the activation functions that follow the linear projections to provide sensible uncertainty estimations.

We show that auNN obtains well-calibrated estimations for in-between data, and its prior allows to guide the extrapolation to OOD data by reverting to the empirical mean. This will be visualized in a simple 1D example (Figure 3 and Table 1). Moreover, auNN obtains competitive performance in standard benchmarks, is scalable (datasets of up to ten millions training points are used), and can be readily used for classification. The use of GPs for the activations establishes an interesting connection with deep GPs (DGPs) (Damianou & Lawrence, 2013; Salimbeni & Deisenroth, 2017). The main difference is the linear projection before the GP, recall Figure 1(c-d). This allows auNN units to model simpler mappings between layers, which are defined along one direction of the input space, similarly to neural networks. However, DGP units model more complex mappings defined on the whole input space, see also Figure 2a. We will show that auNN units require fewer inducing points and are better suited for deep architectures, achieving superior performance. Also, a thorough discussion on additional related work will be provided in Section 4.

In summary, the main contributions of this paper are: (1) a new approach to model uncertainty in DNNs, based on deterministic weights and simple stochastic non-linearities (in principle, not necessarily modelled by GPs); (2) the specific use of non-parametric GPs as a prior, including the triangular kernel inspired by the ReLU; (3) auNN addresses a well-known limitation of BNNs and fBNNs (uncertainty underestimation for in-between data), can guide the extrapolation to OOD data by reverting to the empirical mean, and is competitive in standard prediction tasks; (4) auNN units require fewer inducing points and are better suited for deep architectures than DGP ones, achieving superior performance.

2 PROBABILISTIC MODEL AND INFERENCE

Model specification. We focus on a supervised task (e.g. regression or classification) with training data² $\{\mathbf{x}_{n,:}, \mathbf{y}_{n,:}\}_{n=1}^N$. The graphical model in Figure 2b will be useful throughout this section. We

²The output is represented as a vector since all the derivations apply for the multi-output case.

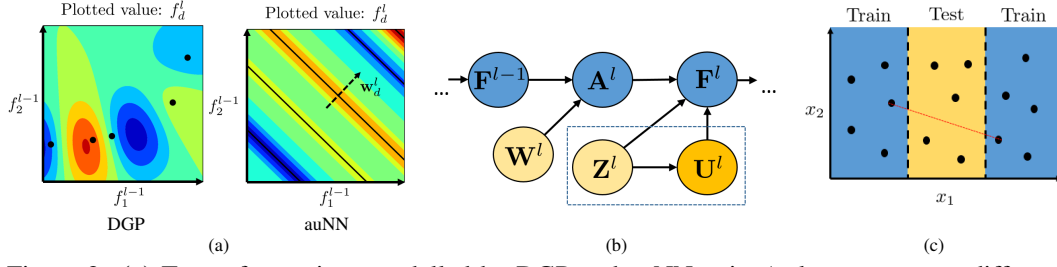


Figure 2: **(a)** Type of mappings modelled by DGP and auNN units (colours represent different values). Whereas DGP units describe complex functions defined on the whole D^{l-1} dimensional input space, the linear projection through $\mathbf{w}_d^l$ in auNN yields simpler functions defined on just one direction. This is closer in spirit to NNs, requires fewer inducing points, and is better suited for deep architectures. The inducing points are shown in black (for auNN, these correspond to (hyper)planes in the input space before the projection). **(b)** Probabilistic graphical model for an auNN layer. Yellow variables are to be estimated (light ones through point estimates and the dark one through a posterior distribution). The box highlights the auxiliary variables (inducing points and their values). **(c)** Graphical representation of the UCI gap splits. In red, a segment that crosses the gap joining two training points from different components, which will be used in the experiments.

assume a model of L layers, each one with D^l units as in Figure 1c. Each activation is modelled with a (1D) GP prior, i.e. $f_d^l(\mathbf{a}_d^l) \sim \mathcal{GP}(\mu_d^l, k_d^l)$, with $\mu_d^l : \mathbb{R} \rightarrow \mathbb{R}$ and $k_d^l : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$. The GP hyperparameters θ_d^l will be omitted for clarity (for the kernels used here, θ_d^l includes the amplitude and the lengthscale). Assuming independence between units, each layer depends on the previous one as:

$$p(\mathbf{F}^l | \mathbf{F}^{l-1}, \mathbf{W}^l) = p(\mathbf{F}^l | \mathbf{A}^l) = \prod_{d=1}^{D^l} p(f_d^l | \mathbf{a}_d^l), \quad (1)$$

where $\mathbf{F}^l$ is the $N \times D^l$ matrix of outputs of the l -th layer for N inputs, $\mathbf{W}^l$ is the $D^{l-1} \times D^l$ matrix of weights in that layer, and $\mathbf{A}^l$ is the $N \times D^l$ matrix of pre-activations, i.e. $\mathbf{A}^l = \mathbf{F}^{l-1} \cdot \mathbf{W}^l$. As usual, the columns and rows of $\mathbf{F}^l$ are denoted as $\mathbf{f}_d^l$ and $\mathbf{f}_{n,:}^l$, respectively (and analogously for the other matrices). Since the activation is defined by a GP, we have $p(\mathbf{f}_d^l | \mathbf{a}_d^l) = \mathcal{N}(\mathbf{f}_d^l | \mu_d^l, \mathbf{K}_d^l)$, with μ_d^l (resp. $\mathbf{K}_d^l$) the result of evaluating μ_d^l (resp. k_d^l) on $\mathbf{a}_d^l$ (that is, μ_d^l is a N -dimensional vector and $\mathbf{K}_d^l$ is a $N \times N$ matrix). To fully specify the model, the output $\mathbf{Y}$ is defined from the last layer with a distribution that factorizes across data points, i.e. $p(\mathbf{Y} | \mathbf{F}^L) = \prod_{n=1}^N p(\mathbf{y}_{n,:} | \mathbf{f}_{n,:}^L)$. This formulation resembles that of DGPs (Damianou & Lawrence, 2013; Salimbeni & Deisenroth, 2017). The main difference is that we model $\mathbf{F}^l | \mathbf{F}^{l-1}$ through D^l 1D GPs evaluated on the pre-activations $\mathbf{A}^l$ (i.e. the projections of $\mathbf{F}^{l-1}$ through $\mathbf{W}^l$), whereas DGPs use D^l GPs of dimension D^{l-1} evaluated directly on $\mathbf{F}^{l-1}$, recall Figure 1(c-d).

Variational Inference. Inference in the proposed model is intractable. To address this, we follow standard sparse variational GP approaches (Titsias, 2009; Hensman et al., 2013; 2015), similarly to the Doubly Stochastic Variational Inference (DSVI) for DGPs (Salimbeni & Deisenroth, 2017). Specifically, in each unit of each layer we introduce M^l inducing values $\mathbf{u}_d^l$, which are the result of evaluating the GP on the one-dimensional inducing points $\mathbf{z}_d^l$. We naturally write $\mathbf{U}^l$ and $\mathbf{Z}^l$ for the corresponding $M^l \times D^l$ matrices associated to the l -th layer, respectively. Following eq. (1), the augmented model for one layer is

$$p(\mathbf{F}^l, \mathbf{U}^l | \mathbf{F}^{l-1}, \mathbf{W}^l, \mathbf{Z}^l) = p(\mathbf{F}^l | \mathbf{U}^l, \mathbf{A}^l, \mathbf{Z}^l) p(\mathbf{U}^l | \mathbf{Z}^l) = \prod_{d=1}^{D^l} p(\mathbf{f}_d^l | \mathbf{u}_d^l, \mathbf{a}_d^l, \mathbf{z}_d^l) p(\mathbf{u}_d^l | \mathbf{z}_d^l). \quad (2)$$

Variational inference (VI) involves the approximation of the true posterior $p(\{\mathbf{F}^l, \mathbf{U}^l\}_l | \mathbf{Y})$. Following (Hensman et al., 2013; Salimbeni & Deisenroth, 2017), we propose a posterior given by $p(\mathbf{F} | \mathbf{U})$ and a parametric Gaussian on $\mathbf{U}$:

$$q(\{\mathbf{F}^l, \mathbf{U}^l\}_l) = \prod_{l=1}^L p(\mathbf{F}^l | \mathbf{U}^l, \mathbf{A}^l, \mathbf{Z}^l) q(\mathbf{U}^l) = \prod_{l=1}^L \prod_{d=1}^{D^l} p(\mathbf{f}_d^l | \mathbf{u}_d^l, \mathbf{a}_d^l, \mathbf{z}_d^l) q(\mathbf{u}_d^l), \quad (3)$$

where $q(\mathbf{u}_d^l) = \mathcal{N}(\mathbf{u}_d^l | \mathbf{m}_d^l, \mathbf{S}_d^l)$, with $\mathbf{m}_d^l \in \mathbb{R}^{M^l}$ and $\mathbf{S}_d^l \in \mathbb{R}^{M^l \times M^l}$ variational parameters to be estimated. Minimizing the KL divergence between $q(\{\mathbf{F}^l, \mathbf{U}^l\}_l)$ and the true posterior is equivalent to maximizing the following evidence lower bound (ELBO):

$$\log p(\mathbf{Y} | \{\mathbf{W}^l, \mathbf{Z}^l\}_l) \geq \text{ELBO} = \sum_{n=1}^N \mathbb{E}_{q(\mathbf{f}_{n,:}^L)} [\log p(\mathbf{y}_{n,:} | \mathbf{f}_{n,:}^L)] - \sum_{l=1}^L \sum_{d=1}^{D^l} \text{KL}(q(\mathbf{u}_d^l) || p(\mathbf{u}_d^l)). \quad (4)$$